

Beyond univariate analysis

Predicting species richness with model-based ordination

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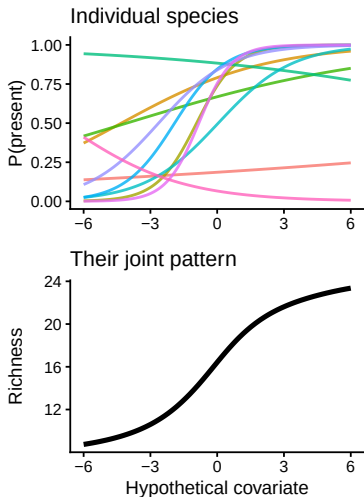
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Modelling the process, not the summary

- ▶ Community ecology cares about **co-occurrence patterns**: which species are found together, and why
- ▶ Richness is a **summary of co-occurrence**: literally a count of how many species co-occur

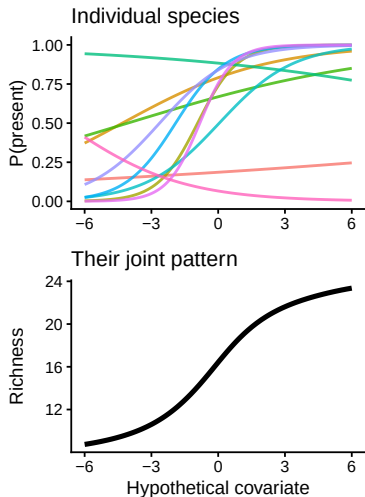
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The standard approach

Species richness is most often analyzed with count regression (O'Hara, 2005) , with $\pi_{ij} = p(y_{ij} > 0)$:

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For this to be successful we require four assumptions that are nearly always violated:

1. All species **respond identically** to the environment
2. Species are **independent** of each other
3. The species pool is **infinite** (Poisson/NB are unbounded)
4. The chosen distribution **is appropriate** for the data

Consequence: the wrong distribution

Heterogeneous niches $\Rightarrow$ richness is **Poisson-Binomial**: **bounded** ($\leq m$) and **underdispersed**.

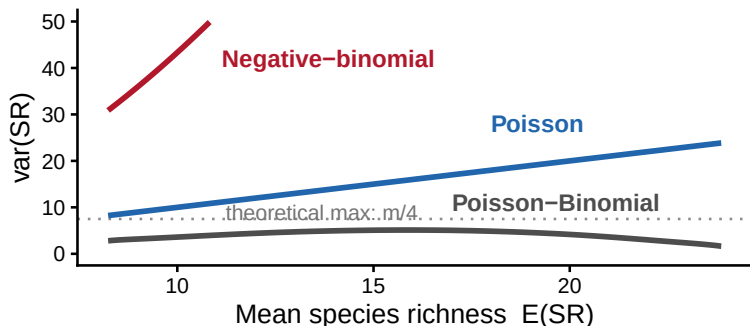
$$\mathbb{E}(SR_i) = \sum_{j=1}^m g_j^{-1}(\mathbf{x}_i^\top \beta_j) \quad (2)$$

Even if each species' response $\mathbf{x}_i^\top \beta_j$ is **linear** in a covariate, each species contributes its own **component-wise** nonlinear curve: the summed richness gradient is effectively **semi-parametric**, not a single bent line.

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$$\mathbb{E}(SR_i) = \sum_j \pi_{ij}, \quad \text{var}(SR_i) = \sum_j \pi_{ij}(1 - \pi_{ij}) \quad (2)$$



Model the community, not the richness

A joint model relaxes the assumptions, and gets the process right, not just the prediction.

Any data type with zero support

$$\mathbb{E}(SR_i) = \sum_{j=1}^m \pi_{ij} \quad (3)$$

- ▶ **Species-specific** responses
- ▶ **Dependence** via ordination
- ▶ **Bounded** by m species

Data type	$\pi_{ij} = p(y_{ij} > 0)$
Binary (Bernoulli)	μ_{ij}
Count (Poisson)	$1 - e^{-\mu_{ij}}$
Ordinal	$1 - p(\text{class } 0)$
Biomass (Tweedie)	$1 - \text{Tw}(0; \mu, \phi, p)$
Cover (ord. beta)	$1 - \rho_{ij}^0$

The community model is a GLLVM

Richness is predicted from each species' response in this model:

$$g\{\mathbb{E}(y_{ij} \mid \mathbf{u}_i)\} = \beta_{0j} + \underbrace{\mathbf{x}_i^\top \beta_j}_{\text{species-specific responses}} + \underbrace{\mathbf{u}_i^\top \gamma_j}_{\text{dependence via ordination}}$$

The latent variables $\mathbf{u}_i$ give model-based ordination:

- ▶ $\mathbf{u}_i = \epsilon_i$: **unconstrained**
(num.lv)
- ▶ $\mathbf{u}_i = \mathbf{B}^\top \mathbf{x}_i$: **constrained**
(num.RR)
- ▶ $\mathbf{u}_i = \mathbf{B}^\top \mathbf{x}_i + \epsilon_i$: **concurrent**
(num.lv.c)

```
gllvm(Y, X, num.lv.c = 2)
```

- ▶ **Facilitate sparse data** by using ordination (fewer parameters)
- ▶ **Incorporate random effects** to accommodate hierarchical study designs
- ▶ **Quantify uncertainty** with confidence intervals
- ▶ **Predict** community composition
- ▶ **Residual diagnostics** for model validation

Predicting richness: predictSR

Once richness is treated as Poisson-Binomial, its **full** distribution is available, not just a mean:

```
sr_pred <- predictSR(model)
sr_pred$expected      # E[SR] with 95% CI
sr_pred$predicted     # full Poisson-Binomial distribution
```

- ▶ $\mathbb{E}(SR_i) = \sum_j \pi_{ij}$: expected richness, with confidence intervals
- ▶ the **mode** in closed form: the most probable integer count
(expected = "mode")
- ▶ $p(SR_i \leq c)$: probability richness is below any threshold (from \$predicted)

Residual diagnostics for richness

```
res <- residuals(sr_pred) # Dunn-Smyth residuals  
plot(sr_pred)            # 3 diagnostic plots
```

- ▶ **Dunn-Smyth** (randomized quantile) residuals, computed from each site's predicted Poisson-Binomial distribution
- ▶ Check the **richness-level** fit: well-behaved residuals and a straight Q-Q line confirm the Poisson-Binomial fit
- ▶ Available for the sites the model was fitted to

Calibrating richness predictions with AUC

Richness is built from many individual presence/absence calls:

$$\mathbb{E}(SR_i) = \sum_j \pi_{ij}$$

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goodnessOfFit(model, measure = "AUC") # conditional on LVs
# latent variables at zero
predict(model, level = 0, type = "response")
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- ▶ If the species-level probabilities **can't discriminate** presence from absence, richness built by summing them inherits that unreliability
- ▶ Complements Dunn-Smyth residuals: those check the **richness-level** fit; AUC quantifies predictive capability

The practical

Data: Skabbholmen (Stockholm archipelago), **ordinal cover classes** along an elevation gradient on a rising shoreline, surveyed in 1978 and 1984

What we will do:

1. Fit a multispecies model with species-specific responses
2. Predict richness with predictSR and check residuals
3. Calibrate with AUC
4. Add a year effect with constrained ordination
5. Compare against a naive Poisson GLM
6. Get composition from the same model
7. Q and A, alternative dataset, whatever you want